D3 EDA: Does Sephora India show up in AI shopping answers?

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> **Decision supported:** Sephora India's marketing and digital team must decide which content and product-feed fixes to make first so that AI shopping assistants name and link Sephora India when shoppers ask for beauty products. **Method:** 20 shopper questions put to 5 AI engines; plain text rules count which retailers are named and linked. 6 hypotheses were written before any chart (page 2).

The five insights that matter most

1. **Nykaa is named about 3.0 times as often as Sephora India.** 32% vs 10% of 222 unbranded answers (53 name only Nykaa, 6 only Sephora; McNemar p<0.001). Nykaa is ahead on 5 of 5 engines and in single fresh chats alone (55% vs 13%).
2. **Sephora appears when shoppers ask where to buy, not what to buy.** 83% of 12 retailer questions vs 6% of 210 product questions (Fisher p<0.001). Fix product-answer visibility first.
3. **Engines differ.** Sephora is named in 22% of ChatGPT, 11% of Perplexity, 9% of Claude, 5% of Gemini, 3% of Google AI Mode answers (chi-square p=0.016). Google AI Mode names Sephora least.
4. **Engines link to Nykaa far more than to Sephora.** Of 74 answers with a retailer link, Nykaa is linked in 74% and Sephora in 23%.
5. **One answer is not reliable.** Cited sites overlap 0.30 (1 = identical) between repeat runs of the same question (Gemini 0.05), so any monitor needs several runs per question.

Figure 4. H1: share of unbranded answers naming each retailer (95% CI)

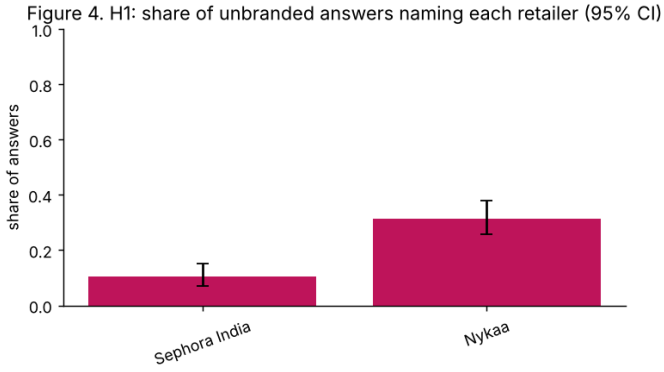
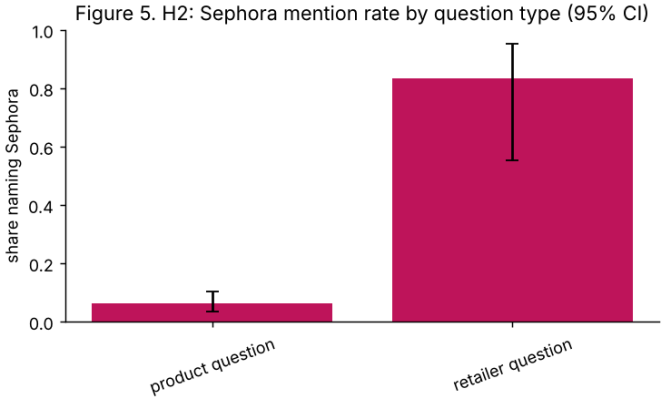


Figure 5. H2: Sephora mention rate by question type (95% CI)



How this shapes modelling (D4)

- Outcome to model: Sephora named (yes/no) per answer. About 10% are positive, so use stratified folds and report precision-recall, not accuracy.
- Inputs: question type, category, engine, collection method, early position in the answer.
- All runs of one prompt stay in the same train or test fold, because repeat runs of a prompt are not independent.
- Only a few retailer-question answers exist, so model product questions first and add where-to-buy when more data arrives.

Hypotheses, bias checks and insight log

Hypotheses stated before analysis, and what we found

#	Hypothesis	Business question	Result
H1	In unbranded prompts, Nykaa is named in a larger share of answers than Sephora India *(expected: Nykaa higher)*	Are we behind the main competitor in AI answers?	Supported: 32% vs 10%, p<0.001
H2	Sephora India is named more often when the shopper asks where to buy (a retailer question) than when they ask for a product (need, budget or shortlist question) *(expected: retailer questions higher)*	Is our visibility a retailer problem or a product problem?	Supported: 83% vs 6% (n=12 vs 210)
H3	Sephora India mention rate differs by AI engine *(expected: differences exist)*	Which engine should we fix first?	Supported: 3% to 22% across 5 engines, p=0.016
H4	Sephora India mention rate differs by category, and is higher for fragrance and premium items than for haircare *(expected: differences exist)*	Where is our strongest and weakest category?	Borderline among product categories (p=0.058); foundation 17% highest, acne_skincare 0% lowest
H5	When an answer links to a retailer, it links to Nykaa more often than to Sephora India *(expected: Nykaa higher)*	Do AI engines send traffic to us or to competitors?	Supported: 74% vs 23%, p<0.001
H6	The same prompt on the same engine gives different cited sources across repeat runs (overlap below 1) *(expected: overlap well below 1)*	Can we trust one answer as the truth, and how many runs do we need?	Supported: mean overlap 0.30 over 68 prompt-engine pairs

Representativeness and bias checks

Figure 1. Planned vs collected share of runs by engine

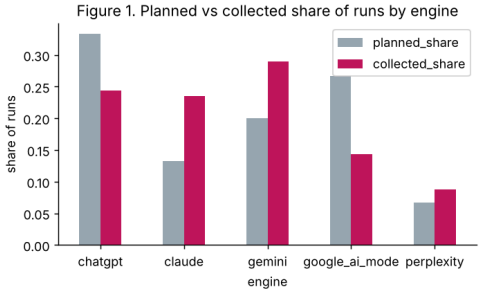
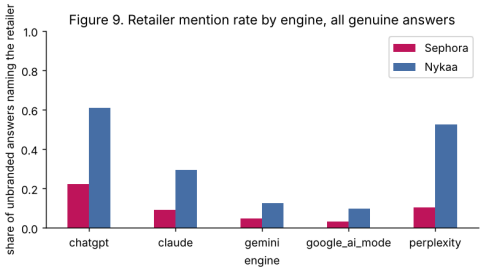


Figure 9. Retailer mention rate by engine, all genuine answers



Engine mix: Gemini 29%, ChatGPT 24%, Claude 24%, Google AI Mode 14%, Perplexity 9% of the 238 answers (plan: popularity-weighted). **Collection method** (141 batched chat, 57 single chat, 40 api) is kept as a variable and tested instead of excluding rows: effect on Sephora mentions none detected (chi-square p=0.67; inside ChatGPT alone p=1.0). **Collector effect:** none detected (p=0.11).

Insight log

#	Insight (evidence)	So what for the decision	Confidence
H1	Sephora named in 10% vs Nykaa 32% of 222 unbranded answers; 53 name only Nykaa, 6 only Sephora (McNemar p<0.001)	Nykaa is named about 3.0 times as often as Sephora India. The first goal is to be named at all in product answers, not to win the retailer comparison.	High: large gap, Nykaa ahead on 5 of 5 engines, also in single fresh chats alone (Fig 9)
H2	Sephora mention rate by question type: product question 6% (n=210), retailer question 83% (n=12) (Fisher p<0.001)	Sephora shows up when shoppers ask WHERE to buy and rarely when they ask WHAT to buy. Fixes should target product-answer visibility (product pages, feeds, reviews), not brand pages.	Medium: only 12 retailer-question answers
H3	Sephora mention rate by engine: chatgpt 22% (n=54), claude 9% (n=54), gemini 5% (n=64), google_ai_mode 3% (n=31), perplexity 11% (n=19) (chi-square p=0.016)	Engines differ. chatgpt names Sephora most; google_ai_mode least. Weight the fixes by engine usage (see the benchmarking note) and re-check on the updated data.	Medium: engine cells are small (19 to 64) and collection method differs by engine
H4	Sephora mention rate by category: acne_skincare 0%, foundation 17%, fragrance 8%, haircare 3%, lipstick 6%, retailer 83%, sunscreen 3%. Product categories only: chi-square p=0.058.	Apart from the retailer category, Sephora is weak across product categories, including its core ones. Treat it as a general product-answer problem until more data shows a category pattern.	Low: 20 to 40 answers per category, wide intervals
H5	Among 74 answers that link to a retailer: Sephora linked in 23%, Nykaa in 74% (McNemar p<0.001)	When engines send shoppers to a retailer, it is usually Nykaa. This is the traffic side of the visibility gap.	Medium: depends on which engines show links at all
H6	Mean overlap of cited sites across repeat runs: 0.30 (chatgpt 0.26, claude 0.36, gemini 0.05, google_ai_mode 0.20, perplexity 0.80) over 68 prompt-engine pairs	One answer is not the truth. Cited sites change between runs, so any monitor needs several runs per prompt.	Medium: few repeat pairs per engine
Bias	Engine mix collected: gemini 29%, chatgpt 24%, claude 24%, google_ai_mode 14%, perplexity 9%. Collection method: batched_chat 141, single_chat 57, api 40.	Mix is closer to the plan than before but Saurav's rows and part of Agnik's Google AI Mode rows are still to come. Results are re-run when the data is updated.	High (a fact about our sample)
Robust	Single fresh chats only (n=53): Sephora 13% vs Nykaa 55% (McNemar p<0.001); all genuine rows: Sephora 10% vs Nykaa 32%	The direction does not depend on the collection method. Collection method has no significant effect on Sephora mentions (p=0.67; inside ChatGPT alone p=1.0), so batched and API answers stay in. Method is kept as a control variable in D4.	Medium-high

Limits: Part submission: the analysis is re-run when the data is updated. Collected with our own accounts; AI answers change with model versions. • Engine mix differs from the popularity weights in the plan; the engine and category cells are small, so p-values there are indicative. • The question-type test rests on a small number of retailer-question answers. • Rules count the retailer name, not the quality of the recommendation. A brand sold at Sephora is not counted as Sephora. • Product-only answers rarely name a retailer, so mention rates are low for every retailer. Compare retailers, not absolute levels. • Gemini API answers and batched chats were collected differently from single chats; the method check shows no significant effect. • No sales or traffic data: we cannot say what a mention is worth in rupees.